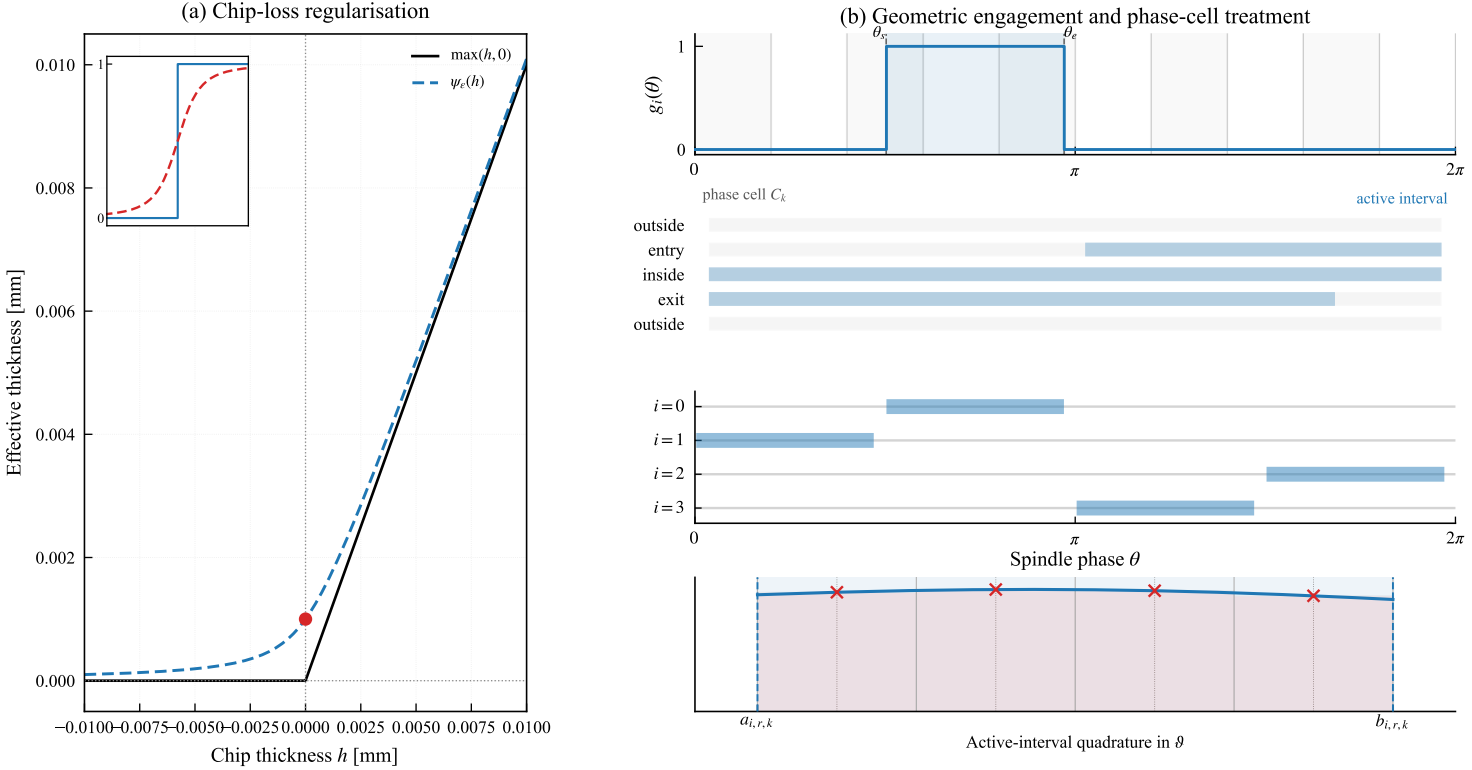

(a) Chip-loss regularisation
max(h, 0)
ψ_ε(h)
Effective thickness [mm]
Chip thickness h [mm]
0.010
0.008
0.006
0.004
0.002
0.000
−0.0100 −0.0075 −0.0050 −0.0025 0.0000 0.0025 0.0050 0.0075 0.0100
(b) Geometric engagement and phase-cell treatment
g_i(θ)
θ_r
θ_e
0
π
2π
phase cell C_k
active interval
outside
entry
inside
exit
outside
i = 0
i = 1
i = 2
i = 3
Spindle phase θ
a_i,r,k
b_i,r,k
Active-interval quadrature in ϑ